

Correlation structure of the **covar** and **orig** data-generating processes

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1 Variable ordering and block partition

Both implementations draw each participant's trajectory from a single multivariate normal over the same ordered variable list. With n measurement occasions at cumulative weeks $w_1 < w_2 < \dots < w_n$, the vector is

$$X = (\text{BM}, \text{BL}, TV_1, \dots, TV_n, PB_1, \dots, PB_n, BR_1, \dots, BR_n)^\top, \quad \dim X = 2 + 3n.$$

The ordering is factor-major: all n occasions of one factor are contiguous, rather than all three factors at one occasion. Writing A_c for the $n \times n$ within-factor block of factor $c \in \{tv, pb, br\}$ and $C_{cc'}$ for the $n \times n$ cross-factor block, the correlation matrix partitions as

$$R = \begin{pmatrix} 1 & 0 & 0^\top & 0^\top & b^\top \\ 0 & 1 & 0^\top & 0^\top & 0^\top \\ 0 & 0 & A_{tv} & C_{tv,pb} & C_{tv,br} \\ 0 & 0 & C_{pb,tv} & A_{pb} & C_{pb,br} \\ b & 0 & C_{br,tv} & C_{br,pb} & A_{br} \end{pmatrix}.$$

Two features hold in both implementations. The baseline term BL is uncorrelated with every other variable, so its row and column are those of the identity. The biomarker BM couples only to the response factor BR, through the vector $b \in \mathbb{R}^n$; its correlations with TV and PB are zero. The two implementations differ in the interior of A_c , in the interior of $C_{cc'}$, and in b .

2 Within-factor blocks A_c

Let ρ_c denote the autocorrelation parameter of factor c (**c.tv**, **c.pb**, **c.br** in the code).

orig: compound symmetry. A single constant is assigned to every pair of occasions, irrespective of their separation:

$$[A_c]_{ij} = \begin{cases} 1, & i = j \\ \rho_c, & i \neq j \end{cases} \iff A_c = (1 - \rho_c)I_n + \rho_c J_n,$$

with J_n the $n \times n$ matrix of ones. For $n = 4$,

$$A_c^{\text{orig}} = \begin{pmatrix} 1 & \rho_c & \rho_c & \rho_c \\ \rho_c & 1 & \rho_c & \rho_c \\ \rho_c & \rho_c & 1 & \rho_c \\ \rho_c & \rho_c & \rho_c & 1 \end{pmatrix}.$$

The block has two distinct eigenvalues, $1 + (n - 1)\rho_c$ once and $1 - \rho_c$ with multiplicity $n - 1$, so positive definiteness requires $-1/(n - 1) < \rho_c < 1$. The upper bound on admissible correlation therefore tightens as occasions are added.

covar: AR(1) in calendar time. The correlation decays with the elapsed time between occasions, using the cumulative week gap rather than the index difference:

$$[A_c]_{ij} = \rho_c^{|w_i - w_j|}.$$

For $n = 4$, writing $d_{ij} = |w_i - w_j|$,

$$A_c^{\text{covar}} = \begin{pmatrix} 1 & \rho_c^{d_{12}} & \rho_c^{d_{13}} & \rho_c^{d_{14}} \\ \rho_c^{d_{12}} & 1 & \rho_c^{d_{23}} & \rho_c^{d_{24}} \\ \rho_c^{d_{13}} & \rho_c^{d_{23}} & 1 & \rho_c^{d_{34}} \\ \rho_c^{d_{14}} & \rho_c^{d_{24}} & \rho_c^{d_{34}} & 1 \end{pmatrix}.$$

Because the exponent uses calendar time, unequally spaced designs produce unequal decay across consecutive occasions. In the Hybrid design, whose occasions fall at weeks 4, 8, 9, 10, 11, 12, 16, 20, consecutive gaps of one week during the discontinuation phase yield much higher adjacent correlations than the four-week gaps at either end.

3 Cross-factor blocks $C_{cc'}$

Let c_1 denote the same-occasion cross-factor correlation (`c.cf1t`) and $c_\times$ the different-occasion cross-factor correlation (`c.cfct`).

orig. Both are constants:

$$[C_{cc'}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times, & i \neq j \end{cases} \iff C_{cc'} = c_\times J_n + (c_1 - c_\times)I_n.$$

covar. The same-occasion entry is unchanged, but the different-occasion entries inherit the AR(1) decay:

$$[C_{cc'}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times \rho^{|w_i - w_j|}, & i \neq j. \end{cases}$$

4 Biomarker coupling vector b

The vector b carries the biomarker-treatment interaction. This is the only place in R where the two implementations differ in mechanism rather than in decay profile.

covar. The entry is gated on the drug indicator and decays with time since discontinuation $t_{sd,p}$ at rate $\lambda = \ln 2/t_{1/2}$:

$$b_p = \begin{cases} c_{bm}, & \text{occasion } p \text{ on drug} \\ c_{bm} e^{-\lambda t_{sd,p}}, & \text{occasion } p \text{ off drug.} \end{cases}$$

orig. The entry is gated on whether the response mean $\mu_{BR,p}$ is nonzero, and takes only two values:

$$b_p = \begin{cases} c_{bm}, & \mu_{BR,p} \neq 0 \\ 0, & \mu_{BR,p} = 0. \end{cases}$$

Because carryover keeps $\mu_{BR,p}$ nonzero after discontinuation, the position of this step depends on the carryover half-life. At $t_{1/2} = 0$ the step coincides with the drug indicator and the two vectors are equal entry for entry. As $t_{1/2}$ grows they separate in opposite directions: the **covar** entries decay toward zero while the **orig** entries rise to c_{bm} . At $t_{1/2} = 1.0$ in the Hybrid design, $b^{\text{orig}} = c_{bm}\mathbf{1}_n$, so the vector carries no on-drug versus off-drug contrast at all.

5 A worked example

To make the three components concrete, take four occasions drawn from the Hybrid design at cumulative weeks $w = (10, 11, 12, 16)$, with the first and fourth on drug and the second and third off it, so that $t_{sd} = (0, 1, 2, 0)$. Set $\rho_{br} = 0.6$, $\rho_{pb} = 0.4$, $c_1 = 0.3$, $c_\times = 0.1$, $c_{bm} = 0.45$, and a carryover half-life of one week, giving $\lambda = \ln 2 \approx 0.693$. The matrix of week gaps is

$$(|w_i - w_j|) = \begin{pmatrix} 0 & 1 & 2 & 6 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 0 & 4 \\ 6 & 5 & 4 & 0 \end{pmatrix}.$$

The unequal spacing matters: occasions one to three are a week apart, while the fourth sits four weeks after the third.

5.1 Within-factor block A_{br}

$$A_{br}^{\text{orig}} = \begin{pmatrix} 1.0000 & 0.6000 & 0.6000 & 0.6000 \\ 0.6000 & 1.0000 & 0.6000 & 0.6000 \\ 0.6000 & 0.6000 & 1.0000 & 0.6000 \\ 0.6000 & 0.6000 & 0.6000 & 1.0000 \end{pmatrix}, \quad A_{br}^{\text{covar}} = \begin{pmatrix} 1.0000 & 0.6000 & 0.3600 & 0.0467 \\ 0.6000 & 1.0000 & 0.6000 & 0.0778 \\ 0.3600 & 0.6000 & 1.0000 & 0.1296 \\ 0.0467 & 0.0778 & 0.1296 & 1.0000 \end{pmatrix}.$$

Under **orig** every pair of occasions carries 0.6, so the week-apart pair (1, 2) and the six-weeks-apart pair (1, 4) are treated as equally associated. Under **covar** the same two pairs carry 0.6 and 0.0467, a factor of thirteen apart. The consequence for the fourth occasion is the striking one: it is nearly independent of the first three under AR(1), and as strongly tied to them as they are to each other under compound symmetry.

5.2 Cross-factor block $C_{tv,pb}$

$$C_{tv,pb}^{\text{orig}} = \begin{pmatrix} 0.3000 & 0.1000 & 0.1000 & 0.1000 \\ 0.1000 & 0.3000 & 0.1000 & 0.1000 \\ 0.1000 & 0.1000 & 0.3000 & 0.1000 \\ 0.1000 & 0.1000 & 0.1000 & 0.3000 \end{pmatrix}, \quad C_{tv,pb}^{\text{covar}} = \begin{pmatrix} 0.3000 & 0.0400 & 0.0160 & 0.0004 \\ 0.0400 & 0.3000 & 0.0400 & 0.0010 \\ 0.0160 & 0.0400 & 0.3000 & 0.0026 \\ 0.0004 & 0.0010 & 0.0026 & 0.3000 \end{pmatrix}.$$

The diagonal, which carries the same-occasion cross-factor correlation c_1 , is identical in both. The off-diagonal entries decay under **covar** and do not under **orig**, and by the fourth occasion the difference is two orders of magnitude.

A caution attaches to the **covar** block. In the implementation the decay rate applied to a cross-factor pair is taken from whichever factor the outer loop is currently visiting, and because each unordered pair is visited in both orders, the later write survives. The rate appearing in $C_{tv,pb}$ is therefore determined by loop order rather than by any modeling decision. The values above use ρ_{pb} on that basis. This is an implementation artifact and not a property of the AR(1) specification.

5.3 Biomarker vector b

At a carryover half-life of one week,

$$b^{\text{covar}} = (0.4500, 0.2250, 0.1125, 0.4500)^\top, \quad b^{\text{orig}} = (0.4500, 0.4500, 0.4500, 0.4500)^\top.$$

The two off-drug entries halve and quarter under **covar**, tracking $e^{-\lambda t_{sd}}$, whereas under **orig** they take the full c_{bm} , because carryover has left the response mean nonzero and the step therefore never fires.

Removing carryover collapses the difference entirely:

$$b^{\text{covar}} = b^{\text{orig}} = (0.4500, 0.0000, 0.0000, 0.4500)^\top \quad (t_{1/2} = 0).$$

This pair of displays is the essential comparison. The vector b is the only channel through which the biomarker-treatment interaction enters the covariance, and an interaction is identified by b varying between on-drug and off-drug occasions. At $t_{1/2} = 0$ the two implementations produce the same vector. At any positive half-life b^{orig} is constant, so it carries no on-drug versus off-drug contrast, while b^{covar} retains one. Across the grid $t_{1/2} \in \{0, 0.5, 1.0\}$ used in this paper, **orig** therefore supplies one cell identical to **covar** and two in which its interaction channel is degenerate.

6 The three data-generating processes

Organized by where each process writes the biomarker-treatment interaction into the multivariate normal (μ, Σ) . All three share the same variable ordering and block partition of Section 1, and differ only in the entries listed.

	mean	covar	orig
Interaction written into	μ	Σ	Σ
Interaction channel	BR mean shift	b vector	b vector
A_c within-factor	AR(1) $\rho_c^{ w_i-w_j }$	AR(1) $\rho_c^{ w_i-w_j }$	compound symmetry, constant ρ_c
$C_{cc'}$ diagonal	c_1	c_1	c_1
$C_{cc'}$ off-diagonal	$c_\times \rho^{ w_i-w_j }$	$c_\times \rho^{ w_i-w_j }$	constant $c_\times$
b on drug	0	c_{bm}	c_{bm}
b off drug	0	$c_{bm} e^{-\lambda t_{sd}}$	0 or c_{bm}
b off-drug profile	n/a	graded	step
b gate	n/a	onDrug , then t_{sd}	$\mu_{BR} \neq 0$
Gate position	n/a	no	yes
carryover-dependent			
μ shift on drug	$c_{bm} z_i \sigma_{BR}$	none	none
μ shift off drug	0 (step)	none	none
Carryover on BR mean	identical line in all three		
BL row and column	identity in all three		
BM-TV, BM-PB	zero in all three		

Three features of this comparison are worth drawing out.

First, **mean** and **covar** share their entire covariance matrix apart from the vector b . Setting b to zero and adding the mean shift converts one into the other, with every within-factor and cross-factor block untouched. This is the decomposition of Appendix A made concrete: the two architectures are not different models but the two places a single joint distribution can carry an interaction.

Second, **orig** differs from **covar** on three rows simultaneously, in A_c , in the off-diagonal of $C_{cc'}$, and in b . No single row can therefore be held responsible for any difference in results obtained under it, and a comparison against **orig** bounds the joint effect of the three differences rather than decomposing them. Separating them would require running the correlation structure and the correlation gate as crossed factors.

Third, only the b row differs in mechanism rather than in decay profile. The A_c and $C_{cc'}$ rows describe the same quantity decaying at different rates, or not decaying at all. The b row describes different conditions being tested to decide whether the interaction is present. That distinction is what makes the gate row consequential: because **orig** tests the response mean rather than the drug indicator, and carryover keeps the response mean nonzero, its gate opens at off-drug occasions precisely when carryover is present, and its interaction channel loses the on-drug versus off-drug contrast on which identification depends.

The final three rows record what all three processes hold in common. The baseline term is uncorrelated with everything, the biomarker couples to no factor other than the response, and the carryover adjustment to the response mean is implemented by the same line of code in every case.

7 Summary of differences

Component	orig	covar
A_c interior	ρ_c	$\rho_c^{ w_i-w_j }$
$C_{cc'}, i = j$	c_1	c_1
$C_{cc'}, i \neq j$	$c_\times$	$c_\times \rho^{ w_i-w_j }$
b_p , on drug	c_{bm}	c_{bm}
b_p , off drug	0 or c_{bm}	$c_{bm} e^{-\lambda t_{sd,p}}$
BL row/column	identity	identity
BM-TV, BM-PB	0	0

The compound-symmetry form constrains ρ_c more tightly than the AR(1) form as n grows, which is the mechanism behind the narrower positive-definite range of c_{bm} under **orig**.